Secure wiring and finish the design

antocciphysics

UNIT 8 / DAY 7 / CPA

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Document permanent connections and evaluate added mass or friction.

SUCCESS CRITERIA - I can...

1. Check circuit continuity and mechanical security.
2. Preserve moving-part clearance during decoration.
3. Relate a design revision to an energy transfer or force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

At unchanged wheel rpm, circumference becomes 4 times as large. Compare ideal travel speed and state one real-world limitation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: I = 0 with power disconnected.
Disconnect before soldering/securing wires. Use insulation and strain relief; route wires clear of gears. Follow supervised tool procedures.

Worked reasoning: A secure circuit also needs strain relief: a pull on a wire should not be carried solely by a solder joint. Insulation prevents exposed conductors from making unintended contacts.

Your turn: Draw one wire route showing a connection, insulation, strain relief and clearance from a rotating axle.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw your final wiring route. Mark soldered joints, insulation, strain relief and the switch location. Identify one possible wheel or gear interference point.



Marzano check - circle: ① ② ③

04 Apply the physics

A model car’s mass increases from 200 g to 250 g after decorations are added. Calculate added mass and fill in the masses in kilograms. Predict how acceleration changes if net forward force stays the same. Explain why that assumption needs checking.

Design	Mass (kg)	Net force (N)	Acceleration
Before	___	___	___
After	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A wire rubs a wheel. Explain the possible effect on energy transfer and propose a repair before another powered test.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed current, move from r to 4 r from a long straight wire. Using B proportional to 1/r, compare field strength. Mark the distances and field directions you compare.

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

07 Exit check

Record one quality check your partner completed and one revision supported by that check.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Create a power-off inspection checklist with three specific observations that would make you stop before powering the car. Explain the concern behind each.



Marzano check - circle: 1 2 3